

W8(2) HARD

SCENARIO PRACTICE

IEEE 802.15.4 + ZigBee

Long questions, hard patterns, traps, design questions & quick drills

WHAT THIS PDF TRAINS

- Scenario -> identify the right technology
- Justify choice with syllabus points
- Comparison under pressure
- Node-failure / routing / security analysis
- Full IoT network design questions
- Short traps + 1-2 mark rapid practice

Exam Target

If time is short, practice Scenarios 1, 2, 4, 5, 7, 8 and 11 first.

Answer pattern: Identify -> Reason -> 5-7 points -> Conclusion.

Scenario 1: Smart Agriculture

Long-question pattern - identify, justify, explain

SCENARIO

A farmer wants hundreds of battery-powered sensors across a field. Each sensor sends only small temperature and soil-moisture readings occasionally. Power use must be very low and wireless communication is required.

QUESTION

Which technology from W8(2) is most suitable? Explain its important features and justify your choice.

Best answer: IEEE 802.15.4

- Designed for low-data-rate WPAN applications.
- Suitable for monitoring and control systems.
- Low power consumption supports long battery life.
- Uses DSSS modulation and CSMA-CA for channel access.
- Low-speed versions use BPSK; high-data-rate versions use O-QPSK.
- Normal transmission range is about 10-75 m; best outdoor case can reach about 1000 m.
- Supports Star and Mesh topologies.

MEMORY: small data + battery + sensor monitoring = IEEE 802.15.4

Scenario 2: Factory Synchronization

Long-question pattern - identify, justify, explain

SCENARIO

A factory has one PAN coordinator and many wireless sensors. Management wants the sensors synchronized with the coordinator and prefers communication in fixed time slots to reduce contention.

QUESTION

Should the network be Beacon-Enabled or Non-Beacon-Enabled? Explain with reasons.

Best answer: Beacon-Enabled Network

- Periodic beacon messages are transmitted.
- Data frames use Slotted CSMA/CA.
- A superframe structure is used.
- The PAN coordinator manages the superframe.
- Beacons provide synchronization.
- Beacons also help association of nodes with the coordinator.
- The operation spans the whole network.

MEMORY: BSS = Beacon - Slotted - Synchronization

Scenario 3: Decentralized Sensor Network

Long-question pattern - identify, justify, explain

SCENARIO

A research team needs a wireless sensor network where devices communicate in a more decentralized way. They do not need periodic synchronization, and each transmission contains source and destination information.

QUESTION

Identify the IEEE 802.15.4 network type and explain how it works.

Best answer: Non-Beacon-Enabled Network

- Uses Unslotted CSMA/CA.
- Communication is contention based.
- Beacons are used mainly for link-layer discovery.
- Both source and destination IDs are required.
- Addressing follows mesh configurations.
- Communication among nodes is decentralized.

MEMORY: USD = Unslotted - Source/Destination IDs - Decentralized

Scenario 4: Smart Building Node Failure

Long-question pattern - identify, justify, explain

SCENARIO

A smart building uses ZigBee. Sensor A normally reaches its destination through node C. Node C suddenly fails, but packets still arrive through other nodes.

QUESTION

Which ZigBee feature explains this behavior? Why does it improve reliability?

Best answer: ZigBee Mesh Networking

- Nodes within range can communicate directly.
- If nodes are out of range, intermediate nodes relay messages.
- Multiple paths can exist between devices.
- If one route or node fails, another route may still carry the packet.
- Mesh increases network reliability.
- Mesh supports deployment over larger areas.
- ZigBee mesh networks are self-configuring and self-healing.

MEMORY: node failed + another path works = Mesh + Self-Healing

Scenario 5: Smart Home Device Roles

Long-question pattern - identify, justify, explain

SCENARIO

A ZigBee smart-home system has three devices. Device A starts the network and stores security information. Device B forwards packets between nodes. Device C is a battery-powered sensor that talks only to its parent and sleeps most of the time.

QUESTION

Identify Device A, B and C and explain their roles.

Best answer: A = ZC, B = ZR, C = ZED

- ZigBee Coordinator (ZC): root of the network; one per network; starts the network; stores network information; acts as Trust Center and stores security keys.
- ZigBee Router (ZR): can run applications and relay information between connected nodes.
- ZigBee End Device (ZED): talks to its parent node, cannot relay other devices' data, can sleep for long periods, and therefore offers long battery life with low memory/cost.

MEMORY: C-R-E = Create network - Relay - End/Sleep

Scenario 6: Battery-Constrained Node

Long-question pattern - identify, justify, explain

SCENARIO

An engineer must choose between FFD and RFD for a tiny battery-powered sensor. The sensor has very little CPU/RAM and only needs to communicate with a more capable device.

QUESTION

Which device type should be selected? Compare it briefly with the alternative.

Best answer: RFD

- RFD can communicate only with an FFD.
- RFD has lower power consumption.
- RFD requires minimal CPU/RAM.
- FFD can communicate with all types of devices.
- FFD supports the full protocol.
- Therefore RFD is the better choice for the constrained sensor.

MEMORY: RFD = Reduced work + Reduced power + Reduced resources

Scenario 7: Unknown Route Discovery

Long-question pattern - identify, justify, explain

SCENARIO

A ZigBee node needs to send data to a destination but does not know the route. It sends a request to nearby nodes, the request spreads through neighbors, the destination is found, and then a low-cost path is returned.

QUESTION

Identify the routing mechanism and explain the sequence.

Best answer: AODV Routing

- AODV means Ad Hoc On-Demand Distance Vector.
- The source broadcasts a Route Request.
- Immediate neighbors receive the request.
- Neighbors relay the request to their neighbors.
- The request spreads until the destination is discovered.
- A low-cost path is calculated.
- Route information is returned to the requester using unicast messaging.

MEMORY: BRFCU = Broadcast - Relay - Find - Calculate - Unicast

Scenario 8: Security and Routing Upgrade

Long-question pattern - identify, justify, explain

SCENARIO

A company already has IEEE 802.15.4 physical and MAC communication. It now needs authentication, encryption, routing, forwarding and mesh capability.

QUESTION

Which technology should be used on top of the lower-layer communication? Explain why.

Best answer: ZigBee

- ZigBee is a widely deployed enhancement of IEEE 802.15.4.
- IEEE 802.15.4 mainly provides the lower layers.
- ZigBee provides higher-layer communication functions.
- It supports authentication of valid nodes.
- It provides encryption for security.
- It adds routing and forwarding.
- It enables mesh networking and is widely used in wireless sensor networks.

MEMORY: AERM = Authentication - Encryption - Routing - Mesh

Scenario 9: Variant Selection

Long-question pattern - identify, justify, explain

SCENARIO

A company has three separate IoT projects: a Smart Grid utility network, an industrial monitoring network, and an Active RFID system.

QUESTION

Select the appropriate IEEE 802.15.4 variants for the three projects.

Best answer: G, E, F

- IEEE 802.15.4g -> Smart utility networks / Smart Grids.
- IEEE 802.15.4e -> Industrial applications.
- IEEE 802.15.4f -> Active RFID uses.
- Other variants from the slides: A/B = Base, C = China, D = Japan.

MEMORY: E = industrial, F = RFID, G = Grid

Scenario 10: Noisy Wireless Environment

Long-question pattern - identify, justify, explain

SCENARIO

An IoT network operates in a noisy wireless environment. The designer wants better tolerance to interference and also wants a channel-access method that checks the medium before transmission.

QUESTION

Which IEEE 802.15.4 mechanisms help satisfy these needs?

Best answer: DSSS + CSMA-CA

- DSSS = Direct Sequence Spread Spectrum.
- DSSS improves tolerance to noise and interference and helps link reliability.
- CSMA-CA = Carrier Sense Multiple Access with Collision Avoidance.
- CSMA-CA is used for channel access and helps reduce collisions.
- Low-speed versions use BPSK.
- High-data-rate versions use O-QPSK.

MEMORY: Noise -> DSSS | Collision avoidance -> CSMA-CA

Scenario 11: Hospital IoT Design

Long-question pattern - identify, justify, explain

SCENARIO

A hospital wants wireless health sensors. Sensors must use little power, some have very low CPU/RAM, communication should be secure, packets may need intermediate relays, and the network should survive some path failures.

QUESTION

Design a suitable solution using W8(2) concepts and justify each choice.

Best answer: IEEE 802.15.4 + ZigBee Mesh

- Use IEEE 802.15.4 as the low-data-rate, low-power base wireless standard.
- Use RFDs for simple constrained sensors because they require little CPU/RAM and use less power.
- Use ZigBee for authentication, encryption, routing and forwarding.
- Use Mesh topology so intermediate nodes can relay data.
- Mesh provides alternative paths, improved reliability and self-healing behavior.
- Use a ZigBee Coordinator to initiate the network, Routers to relay data, and End Devices for low-power sensing.
- Use AODV for on-demand route discovery.

MEMORY: Low power -> 802.15.4 | Security/routing -> ZigBee | Failure tolerance -> Mesh

Scenario 12: Full Campus Architecture

Hard integrated design question - combine the whole chapter

HARD DESIGN

A university campus wants smart lights, energy meters and environmental sensors. Some devices are far apart, many are battery-powered, devices may occasionally fail, and security plus routing are required.

Question: Design an appropriate architecture using only the W8(2) concepts.

1. Base wireless IEEE 802.15.4 - low-data-rate, low-power, PHY/MAC, DSSS, CSMA-CA.

2. Upper functions ZigBee - authentication, encryption, routing, forwarding, mesh support.

3. Topology Mesh - relays through intermediate nodes, large-area coverage, self-healing.

4. Device roles ZC starts network; ZR relays; ZED is low-power end sensor.

5. Routing AODV - route request broadcast, relay, destination discovery, low-cost path, unicast return.

Conclusion line: IEEE 802.15.4 + ZigBee Mesh is a strong fit for this low-power, secure, reliable IoT design.

Hard Trap Questions

True/False + one-line correction

Q1 ZigBee and IEEE 802.15.4 are exactly the same technology.

Answer: FALSE - ZigBee adds higher-layer functions over IEEE 802.15.4 lower layers.

Q2 Every ZigBee device can relay packets.

Answer: FALSE - ZED cannot relay other devices' data.

Q3 RFD can communicate with every device type.

Answer: FALSE - RFD communicates only with an FFD.

Q4 Beacon-Enabled networking uses Unslotted CSMA/CA.

Answer: FALSE - Beacon uses Slotted CSMA/CA.

Q5 A ZigBee mesh must stop if one intermediate node fails.

Answer: FALSE - mesh can use alternative paths and is self-healing.

Q6 FFD is the best choice for a tiny low-RAM battery sensor.

Answer: FALSE - RFD is more suitable when full protocol is not needed.

Rapid Practice - 25 Short Questions

Try without looking at the answers first

1. What is WPAN?
2. What is FFD?
3. What is RFD?
4. Which device supports the full protocol?
5. Which device type has lower CPU/RAM needs?
6. What is DSSS?
7. What is CSMA-CA?
8. Which modulation is used for low speed?
9. Which modulation is used for high data rate?
10. What is the normal IEEE 802.15.4 range?
11. What is the best outdoor range stated in the slides?
12. What is the approximate duty cycle condition?
13. Name two IEEE 802.15.4 topologies.
14. What is ZC?
15. What is ZR?
16. What is ZED?
17. Can ZED relay packets from other devices?
18. What is AODV?
19. What does self-healing mean in mesh networking?
20. What is ZDO?
21. What is APS?
22. Which variant is used for industrial applications?
23. Which variant is used for Active RFID?
24. Which variant is used for Smart Grid?
25. Name four ZigBee applications.

Final Exam Strategy

How to answer unseen scenario questions

MASTER DECISION MAP

Small data + low power	->	IEEE 802.15.4
Synchronization + time slots	->	Beacon Enabled
Decentralized + no periodic sync	->	Non-Beacon Enabled
Security + routing + forwarding	->	ZigBee
Node failure + alternate path	->	Mesh / Self-healing
Low RAM/CPU sensor	->	RFD or ZED (depending on context)
Unknown route	->	AODV

8-MARK ANSWER FORMULA

- 1. Identify the technology/device/network type immediately.
- 2. Write one-line definition.
- 3. Connect scenario clues to the technology.
- 4. Write 5-7 syllabus-supported points.
- 5. Add one exact keyword/number where useful.
- 6. End with one clear conclusion sentence.

Tonight's top 8 practice:

1 Agriculture | 2 Beacon Factory | 4 Mesh Failure | 5 Device Roles | 7 AODV | 8 ZigBee Upgrade | 11 Hospital Design | 12 Campus Design